

One-Dimensional Bin-Packing Problems with Branch and Bound Algorithm

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Abstract: In this paper, our objective is to develop a mathematical formulation of solving the Bin Packing Problem (BPP) with different sizes of box. It has become one of the most important mathematical applications throughout the time. In our study, Modified Branch and Bound Algorithm (MBBA) is developed to generate all the feasible packing patterns of given boxes to required containers for One-Dimensional BPP. Further development of algorithms was made to ascertain the locations of each box within the containers by using Cartesian coordinate system. Developed algorithms are coded and programmed in the Python programming environment to generate feasible packing patterns.

Keywords: BPP, MBBA, Python Software Package, Cartesian Coordinate Points

1. Introduction

Due to manufacturing uprising, at present most of the industries are focusing in the frame of globalization. In turn, industries have substantially augmented the capacity and magnitude of their global manufacture and allocation network around the world. Thus the globalized market results in rapid development of international trade and creates intensive competitions among the industries. In this research, our objective is to investigate a mathematical formulation of solving the One-Dimensional BPP which has been studied by many researchers in the recent past. Packing the produced goods in the optimum manner within a limited space has multiple paybacks to the production plant. This will eventually decrease the production of the plant. A Bin Packing Problem basically describes in three ways; One-Dimensional, Two-Dimensional and Three-Dimensional Bin Packing problems. The Bin Packing problem can be defined as a finite collection of items with varying specifications to be packed into one or more containers utilizing the maximum volume of the containers while satisfying the supply-demand. Each container can hold any subset of the collection of objects without exceeding its capacity. Bin packing is also called as container loading, box packing, cargo loading, knapsack, etc.

A burning issue faced by the industries is how to find the optimum layout (packing arrangement) of boxes or packing items which have different shapes and sizes within the available different sizes of bins (container) such that improve the utilization ratio of bins or minimize the bin slack without overlapping the packages. Jatinder N. D. Gupta and Johnny C. (1999) have described a new Heuristic Algorithm to solve the one-dimensional Bin Packing problem. Effectiveness of the proposed algorithm has been compared with the First Fit Decreasing (FFD) and the Best Fit Decreasing (BFD) algorithms using five different data sets [1]. Solve the one dimensional BPP with island parallel grouping genetic algorithms was the main motivation of Dokeroglu T. et al (2014). Combining state-of-the-art computation tools; parallel processing, GGAs and bin oriented heuristics to efficiently solve the intractable one-dimensional BPP. Different size case studies discussed in the paper using Minimum Bin Slack (MBS) and Best Fit Decrease (BFD) / First Fit Decrease (FFD). [2]. Mukhacheva E. A et al (2000) developed two algorithms for one-dimensional cutting-packing problem, namely, a modified Branch-and-Bound method (MBB) and Heuristic Sequential Value Correction (SVC) method. Efficiency of the algorithms is discussed from the computational experiment and it seems that the efficiency of the SVC method appears to be superior to that of the MBB [3]. Fleszar K. and Khalil S. presented a new

Heuristic Algorithm for minimum bin slack. In this research, each packing is determined in a search procedure that tests all possible subsets of items on the list which $\leq$ the bin capacity (C). The slack in packing A is expressed by $S(A) = C$ and updating every time an item is added or removed from A [4]. Andrea Lodi *et al* (2002) have emphasized on exact algorithms and effective heuristics and meta heuristic approaches to the two-dimensional Bin Packing problem [5].

Christian Blum and Verena Schmid (2013) have dealt with two-dimensional Bin Packing problem under free guillotine cutting, a problem in which a set of oriented rectangular items are given which must be packed into minimum number of bins of equal size. An evolutionary algorithm has been discussed and the results of the proposed algorithm are compared with some of the best approaches from the literature [6]. There is intercommunication in between Cutting Stock Problem (CSP) and BPP. Many researchers have worked on the cutting stock problem as well and designed different algorithmic approaches to solve the problem. Among them, Saad (2001) modified Branch and Bound Algorithm to find feasible cutting patterns for one-dimensional cutting stock problem and formulated a mathematical model to minimize the total cut loss. Modified Branch and Bound Algorithm is illustrated using a case study [7]. Further Rodrigo *et al* (2012) developed an algorithm, based on Modified Branch and Bound algorithm [7] to determine the feasible cutting patterns for Two-Dimensional cutting stock problem with rectangular shape cutting items. The method was illustrated with the use of a case study, where the data were obtained from a floor tile company known as Mega Marble Company located in London. A computer programme was coded using Matlab inbuilt functions [8]. As an extension of the above study, Rodrigo *et al* (2012) redesigned the developed algorithm [8] to determine the locations of each cutting item using Cartesian Coordinate Geometry [9]. Rodrigo *et al* (2015) modified the Branch and Bound Algorithm to decide the feasible cutting patterns for One-Dimensional cutting stock problem and to determine the locations of each cutting item using Cartesian Coordinate Geometry. The algorithm has been coded using Matlab environment illustrated the algorithm using a case study. [10]. Thereafter, Rodrigo *et al* (2017) has made an approach to nest one-dimensional cutting items within rectangular shaped main sheet with known varying dimensions. Cartesian coordinate points of each item in each cutting sheet have been determined using a computer programme coded in Matlab software environment [11]. Besides, identification of cutting-packing location within the different sizes of container (or main sheet) is significantly crucial to outline items within the container depending on the selected cutting-packing pattern. To address this issue, further development of MBBA will be made to ascertain the locations of each cutting-packing item within the main sheet (or container) by using Cartesian coordinate system. Developed algorithm will be coded and programmed in the Python Software environment to generate feasible packing patterns for One-Dimensional BPP.

2 Materials and Method

2.1. One-Dimensional (1D) BPP

In most industries, the cost of the transportation consists of a high percent of total price. The BPP is one of the most famous operation research problems which are defined to improve the amount of packing boxes in to containers. This BPP can be described as follow; there is a container with the rectangular shape base with the width W and the length L . At the same time there are different size of boxes with the rectangular shape base with the width w_i ; $i = 1, 2, \dots, n$ and the length L . In this paper, MBBA applied to generate feasible packing patterns of n number of boxes to bunch into the container.

2.2. Mathematical Model (Gilmore and Gomary, 1961)

Any firm's main objective is to maximize the annual contribution margin accruing from its production and sales. By reducing costs, wastages and maximizing sales, productivity can be improved. Cost can occur in many ways and transportation cost takes a big challenge in industry. According to the selection, a mathematical model to minimize the packing cost is formulated as follows:

$$\text{Minimize } z = \sum_{j=1}^n c_j x_j \quad (\text{Total bin slack})$$

$$\text{Subject to } \sum_j a_{ij} x_{ij} \geq N_j \quad \text{for } i = 1, 2, \dots, m$$

$$x_{ij} \geq 0 \text{ and Integer for all } i, j.$$

Where

n = Number of boxes with different sizes,

m = Number of patterns,

a_{ij} = The number of occurrences of the i^{th} size box in the j^{th} pattern,

c_{kj} = Bin slack of the k^{th} bin in the j^{th} pattern,

N_j = Number of boxes of the i^{th} size box,

x_{ijk} = The number of boxes of i^{th} size box can be packed in the j^{th} pattern.

2.3. Modified Branch and Bound Algorithm (MBBA)

Step 1: Arrange required widths (or lengths) of boxes, w_j , $j = 1, 2, \dots, n$ in decreasing order, ie $w_1, w_2, \dots, w_n$, where n = number of boxes from each sizes.

Step 2: For $i = 1, 2, \dots, m$ and $j = 1$ do Steps 3 to 5.

Step 3: Set

$$a_{11} = \left\lfloor \frac{W}{w_1} \right\rfloor; \quad (1)$$

$$a_{ij} = \left\lfloor \frac{\left(W - \sum_{z=1}^{i-1} a_{zj} w_z \right)}{w_i} \right\rfloor; \quad (2)$$

where

W_k is the width of the container. Here, a_{ikj} is the number of quantity of the i^{th} box in the j^{th} pattern along the width of the container and $[[y]]$ is the greatest integer less than or equal to y .

Step 4: Set $P_j = a_{ij}$

where P_j is the number of quantities of the i^{th} box in the j^{th} pattern within the base of the container.

Location of the i^{th} item in the j^{th} pattern and number of pieces from each item: If $a_{ij} > 0$, then set

$$(x_i, y_i) = \left(\left(\sum_{z=1}^i a_{zj} w_z - a_{ikj} w_i \right), 0 \right); \left(\left(\sum_{z=1}^i a_{zj} w_z \right), 0 \right)$$

$$N_i = a_{ij}$$

where

N_i is the number of pieces can be nest from the i^{th} box in the j^{th} pattern along the width-wise and (x_i, y_i) , are the coordinates of the each box of i^{th} item in the j^{th} pattern within the container.

Step 5: Bin slack along the width of the container:

$$C_j = \left(W - \sum_{i=1}^m a_{ij} w_i \right);$$

Step 6: Set $t = m-1$.

While $t \geq 0$, do Step 7.

Step 7: While $a_{tj} \geq 0$ set $j = j + 1$ and do Step 8.

Step 8: Generate a new pattern according to the following conditions:

For $z = 1, 2, \dots, t-1$;

Set $a_{zj} = a_{z(j-1)}$

For $z = t$

Set $a_{zj} = (a_{z(j-1)} - 1)$

For $z = t + 1, \dots, m$

calculate a_{zj} using (1).

Go to Step 4.

Step 9: Set $t = t-1$.

Step 10: STOP.

2.4. Case Study

Proposed MBBA to solve one-dimensional BPP is tested and analyzed to determine feasible and optimal packing patterns. Following examples will illustrate how to generate feasible packing patterns by minimizing the total bin slack.

Following data table represents two one-dimensional bin packing problem given in the paper [2] and proposed algorithm in this paper has been applied to solve those problems.

Table 1. Required widths of the boxes and containers.

Problem No	Width of the boxes	Width of the base of the container
1	1, 2, 3, 3, 4, 4, 5, 5, 6, 7	10
2	12, 13, 19, 21, 25, 25, 25, 28, 39, 40, 41, 47, 48, 54, 56, 67, 70, 70, 82, 85	100

3. Results and Discussion

Modified Branch and Bound Algorithm is applied to the above examples to generate feasible packing patterns as given below:

Problem No 1 in the Table 1:

Table 2. Required widths and demand of problem 1.

Width of the boxes	Demand
1	1
2	1
3	2
4	2
5	2
6	1
7	1

Following table represent the generated feasible packing patterns applying MBBA for problem 1 given in the table 3. Python programming language is used to code the MBBA.

Table 3. Generated Packing Patterns of problem 1.

Pattern No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Width of the boxes																																	
7	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	2	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	1	0	0	0	0	1	0	0	0	0	2	2	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
3	1	0	0	0	1	0	0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	3	2	2	2	1	1	0	0	0	0	0	0
2	0	1	0	0	0	2	0	0	0	1	0	2	0	1	0	1	0	3	2	1	0	0	2	1	0	3	2	5	4	3	2	1	0
1	0	1	3	0	1	0	4	0	1	0	2	1	5	0	2	1	3	0	2	4	6	1	0	2	4	1	3	0	2	4	6	8	10
Bin Slack	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The mathematical model is solved as an integer programming model using EXCEL to design generated

packing patterns so that the bin slack will be minimized and optimum solution to the model is given in the following table.

Table 4. Optimum solution for problem no 1.

Optimum Pattern No	1	5	8	14	Demand
Width of the boxes					
7	1	0	0	0	1
6	0	1	0	0	1
5	0	0	2	0	2
4	0	0	0	2	2
3	1	1	0	0	2
2	0	0	0	1	1
1	0	1	0	0	1
No of Containers	1	1	1	1	

Cartesian Coordinates obtained from the MBBA:

Pattern no 1: (0, 0), (0, 7); (0, 7), (0, 10)

Pattern no 5: (0, 0), (0, 6); (0, 6), (0, 9); (0, 9), (0, 10)

Pattern no 8: (0, 0), (0, 10)

Pattern no 14: (0, 0), (0, 8); (0, 8), (0, 10)

Similar procedure can be applied for the other problem as given below:

Problem No 2 in the Table 1:

There are 9 different packing patterns according to the MBBA and two containers should be required to satisfy the demand of boxes.

Table 5. Optimum solution for problem no 2.

Optimum	Width of the boxes																	Bin	No of
Pattern No	12	13	19	21	25	28	39	40	41	47	48	54	56	67	70	82	85	Slack	Containers
1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	1
4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	6	1
5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	2	1
6	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	5	1
13	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	8	1
20	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	3	1
33	0	0	1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	2	1
45	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	5	1
57	0	0	0	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1
Demand	1	1	1	1	3	1	1	1	1	1	1	1	1	1	2	1	1		

Cartesian Coordinates points obtained from the MBBA:

Pattern no 1: (0, 0), (0, 85); (0, 85), (0, 98)

Pattern no 4: (0, 0), (0, 82); (0, 82), (0, 94)

Pattern no 5: (0, 0), (0, 70); (0, 70), (0, 98)

Pattern no 6: (0, 0), (0, 70); (0, 70), (0, 95)

Pattern no 13: (0, 0), (0, 67); (0, 67), (0, 92)

Pattern no 20: (0, 0), (0, 56); (0, 56), (0, 97)

Pattern no 33: (0, 0), (0, 54); (0, 54), (0, 79); (0, 79), (0, 98)

Pattern no 45: (0, 0), (0, 48); (0, 48), (0, 95)

Pattern no 57: (0, 0), (0, 40); (0, 40), (0, 79); (0, 79), (0, 100)

Effectiveness of the proposed MBBA has been compared with algorithms given in the paper [2].

Table 6. No of containers for problems given in the paper [2].

No of Containers from different algorithms	MBBA	MBS	BFD/FFD
Problem No			
1	4	4	
2	9		10

4. Conclusion

In this paper, a bin packing problem is formulated as a mathematical model based on the concept of packing patterns. Five different problems are solved using proposed MBBA and results (minimum number of containers to satisfy the demand of each box) are compared with three different algorithms; Minimum Bin slack (MBS), First Fit Decreasing (FFD) and Best Fit Decreasing (BFD). According to the MBS algorithm [2], Minimum containers are found to satisfy the demand of each size of boxes. In our approach; MBBA gives

comparable solution with MBS but dissimilar with BFD/FFD. However, looking the solution of the MBS and BFD/FFD, it's difficult to identify the places of each box should be allocated to into container satisfying the minimum number of containers for each problem. As a result, identification of packing location within the base of the container is significantly vital to trace boxes within the container depending on the selected packing pattern. Accordingly, MBBA can be addressing this issue; to ascertain the locations of each packing item within container by using Cartesian coordinate system. Developed algorithm (MBBA) is coded and programmed in the Python programming environment to generate feasible packing patterns and location of each box in the pattern.

References

- [1] Jatinder N. D Gupta and Johnny C. "A new heuristic algorithm for the one-dimensional bin-packing problem", Production planning and Control, ISSN 0953-7287(1999), Vol. 10, No. 6, 598-603.
- [2] T. Dokeroglu, A. Cosar, "Optimization of one-dimensional Bin Packing Problem with island parallel grouping genetic algorithms" Computers & Industrial Engineering 75 (2014) 176-186.
- [3] Mukhacheva E. A, G. N Belov, V. M. Kartack and Mukhacheva A. S. "Linear one-dimensional cutting-packing problems: Numerical experiments with the sequential value, Correction Method and a modified Branch and Bound method".
- [4] Fleszar K. and Khalil S. "New Heuristics for one-dimensional bin packing", Research-Gate: Computers and Operations Research, DOI: 10.1016/S0305-0548(00)00082-4.

- [5] Andrea L, Silvano M and Daniele V. “*Recent Advances on two-dimensional bin packing Problems*”, Elsevier Science, Discrete Applied Mathematics. 123 (2002), 379-396.
- [6] Christian B. and Verena S. “*Solving the 2D bin packing problem by means of a Hybrid Evolutionary Algorithm*”, Elsevier, International Conference on Computational Science, ICCS (2013), 899-908.
- [7] Saad M. A. Suliman. “*Pattern generating procedure for the cutting stock problem*”, International Journal of Production Economics 74 (2001) 293-301.
- [8] W. N. P. Rodrigo, W. B. Daundasekara and A. A. I. Perera. “*Pattern Generation for Two-Dimensional Cutting Stock Problem*”, International Journal of Mathematics Trends and Technology, Vol. 3, Issue 2: 54-62.
- [9] W. N. P. Rodrigo, W. B. Daundasekara and A. A. I. Perera. “*Pattern Generation for Two-Dimensional Cutting Stock Problem with Location*”, Indian Journal of Computer Science and Engineering (IJCSE), Vol. 3, No 2, April-May 2012, 354-368 (www.academia.edu/4463690/INDJCSE12-03-02-082).
- [10] W. N. P. Rodrigo, W. B. Daundasekara and A. A. I. Perera. “*Modified Method for One-Dimensional Cutting Stock Problem*”, Software Engineering. Vol. 3, No. 3, 2015, pp. 12- 17. doi: 10.11648/j.se.20150303.11.
- [11] Niluka Rodrigo, Sium Shashikala. “*One-Dimensional Cutting Stock Problem with Cartesian Coordinate Points*”, International Journal of Systems Science and Applied Mathematics. Vol. 2, No. 5, 2017, pp. 99-104. doi: 10.11648/j.ijssam.20170205.14.